Newcastle Business School
The University of Newcastle

BUSA3003 – Business Process Modeling
Assignment 2 - BPM Implementation Plan

Smart Travel Ecosystem Project - TripMate

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Executive summary

Tourists often face fragmented travel planning, using multiple apps for flights, accommodation, food, transport, and local activities. This disjointed process is time-consuming, mentally exhausting, and can cause anxiety, uncertainty, and missed opportunities to experience authentic local culture. To overcome these issues, this project developed TripMate, an all-in-one, web-based platform integrating AI-powered itinerary planning, customised suggestions, and seamless booking across all aspects of a trip.

The project utilised a design-thinking model, following the empathise, define, and ideate steps. The Lotus Blossom technique was used to ideate and generated eight interrelated modules: Accommodation, Food & Restaurant, Events and Entertainment, Transportation and Navigation, Local Experience, Budget Management, Social and Sharing, and Smart Recommendation.

A high-fidelity Figma prototype demonstrates how TripMate consolidates planning, booking, and navigation into a single platform. By reducing decision fatigue, saving time, and enhancing deeper involvement in local culture, TripMate will provide a measurable value to travellers, local businesses, and event organisers and guarantee scalability and further optimization in the lifecycle of BPM.

1. Introduction

Tourists often face challenges that reduce the quality of their travel experiences. Itinerary creation is generally fragmented, with flights, accommodation, attractions, and ticket details spread across multiple platforms. This forces tourists to spend excessive time organising each part of their trip, resulting in stress, confusion, and missed opportunities. Access to authentic local experiences is also limited, as cultural events, festivals, and niche activities are often overlooked by mainstream tools, limiting engagement and reducing satisfaction.

To alleviate these shortcomings, this project introduces TripMate, a web-based application integrating AI-driven itinerary planning with personalised, location-aware recommendations. By consolidating essential travel functions into a single intuitive interface, TripMate streamlines planning, supports informed decision-making, and encourages meaningful local engagement. The platform benefits both travellers and stakeholders, fostering efficient, enjoyable, and sustainable tourism experiences.

1. Method - Design Thinking Approach

1.1. Empathise

Secondary research shows that the hospitality and tourism industries have been transformed by technologies such as mobile cloud computing, the Internet of Things (IoT), artificial intelligence (AI), and blockchain. These innovations have enabled Mobile Travel Applications (MTAs) that support digitalised travel experiences (Sia et al., 2023). However, most existing MTAs focus on specific functions—such as flight booking, accommodation, or event discovery—forcing travellers to use multiple apps. Studies indicate that users often download and test numerous tourism apps, which is time-consuming and mentally exhausting (Garcia-Lopez et al., 2021).

To validate these insights, informal interviews and discussions were conducted with friends, colleagues, and regular travellers. Most participants reported relying on multiple platforms—Google Maps or TripViewLite for transport, Airbnb or Booking.com for accommodation, and social media for events and dining. This fragmented approach caused confusion, fatigue, and anxiety, as users struggled to coordinate bookings and feared missing deals. Many expressed a need for a single, intelligent solution that simplifies planning and increases confidence in their travel arrangements.

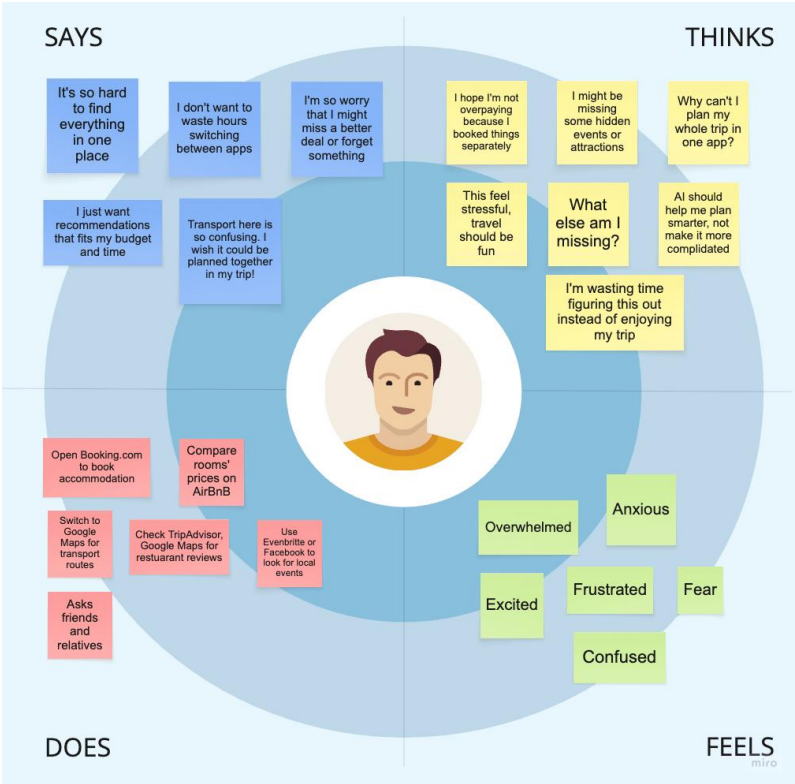


Figure 1. Empathy Map

1.2. Define

Insights from the empathise stage showed that travellers often experience fatigue, anxiety, and reduced motivation during pre-trip planning. The excitement of travel diminishes as users manage multiple apps for flights, accommodation, transport, dining, and events. This fragmented process leads to decision fatigue, inefficiency, and uncertainty, highlighting the need for a cohesive solution that ensures plans are safe, efficient, and cost-effective.

Mapping these findings to Maslow's hierarchy reveals that challenges extend beyond functionality. Physiological and safety needs include secure lodging, reliable transport, and transparent budgeting. Love and belonging are expressed through connecting with locals, participating in cultural experiences, and sharing memories. Self-actualisation is achieved via authentic engagement, learning, and adventure.

The problem statement is clear: travellers need an integrated, intelligent platform that streamlines planning, meets safety and organisational needs, and supports emotional and aspirational fulfilment.

Figure 2. Maslow's hierarchy of needs

1.3. Ideate

The ideate stage focused on generating innovative, user-centred solutions to address the challenges identified during the empathise and define stages. To systematically explore creative possibilities, the Lotus Blossom technique was employed. This method allows designers to expand a central concept into detailed subcategories, ensuring a holistic consideration of the user journey and uncovering potential features that meet functional, emotional, and aspirational needs.

The central concept of the Lotus Blossom diagram was the "All-in-One Tourism Journey App," a unified digital platform designed to integrate all stages of a traveller's experience, from planning and booking to local exploration and sharing. Eight key subcategories were identified around this core idea: Accommodation, Food & Restaurant, Events & Entertainment, Transport & Navigation, Local Experience, Budget Management, Social & Sharing, and Smart Recommendation. Each subcategory was expanded into specific features aligned with user motivations, derived from empathy mapping and Maslow's hierarchy of needs.

The Accommodation module offers hotel, motel, and hostel bookings with integrated user reviews, ratings, and price comparisons. A map view visualises locations, while promotions and package deals address safety and budget considerations. Food & Restaurant provides curated dining options, including street food and restaurants, with dietary filters and integration with platforms like Google Reviews and TripAdvisor. Reservation discounts further enhance user satisfaction.

Events & Entertainment highlights local festivals, cultural activities, and seasonal events, with curated lists for families, couples, or nightlife seekers. Real-time event updates encourage spontaneous exploration and

fulfil social and belonging needs. Transport & Navigation integrates flight bookings, public transport planning via Google Maps or TripViewLite, and bike or scooter rental suggestions, with real-time traffic updates to ensure efficiency and safety.

The Smart Recommendation module employs AI to generate personalised trip plans based on user preferences, past behaviour, and travel type—backpacker, family, luxury, or business—supporting self-actualisation. Local Experience showcases hidden gems and authentic attractions suggested by locals, using Eventbrite and TikTok for engagement. Budget Management includes loyalty cards, in-app wallets, currency converters, and special discounts for efficient financial control. Social & Sharing allows posting of reviews, social media sharing, and connecting with other travellers, fostering belonging, esteem, and community engagement.

Figure 3. Lotus Blossom Diagram

3. Results and Implications

3.2. Prototype Description

3.2.1. AS-IS process and TO-BE process

a. AS-IS process

The AS-IS travel planning process represents the current fragmented approach where tourists use multiple independent platforms to organise their trips, including flights, accommodation, transport, events, and budget management. After deciding on a destination and travel date, travellers must manually search, compare, and book each service separately across different apps or websites. For every booking, users repeatedly enter personal and payment details, receive individual confirmations, and manually track all arrangements before their trip can begin.

This current approach creates several pain points. Repetitive several manual input forces tourists to repeatedly enter destination, date, and budget information across multiple apps, wasting time, increasing fatigue and increasing the risk of data leakage, financial fraud, and loss of privacy. The experience is fragmented, as bookings for flights, hotels, and activities are managed separately, requiring users to track multiple confirmations, which can be particularly confusing for less tech-savvy individuals or accessibility needs. There is no centralized budget control, making it easy to exceed planned spending.

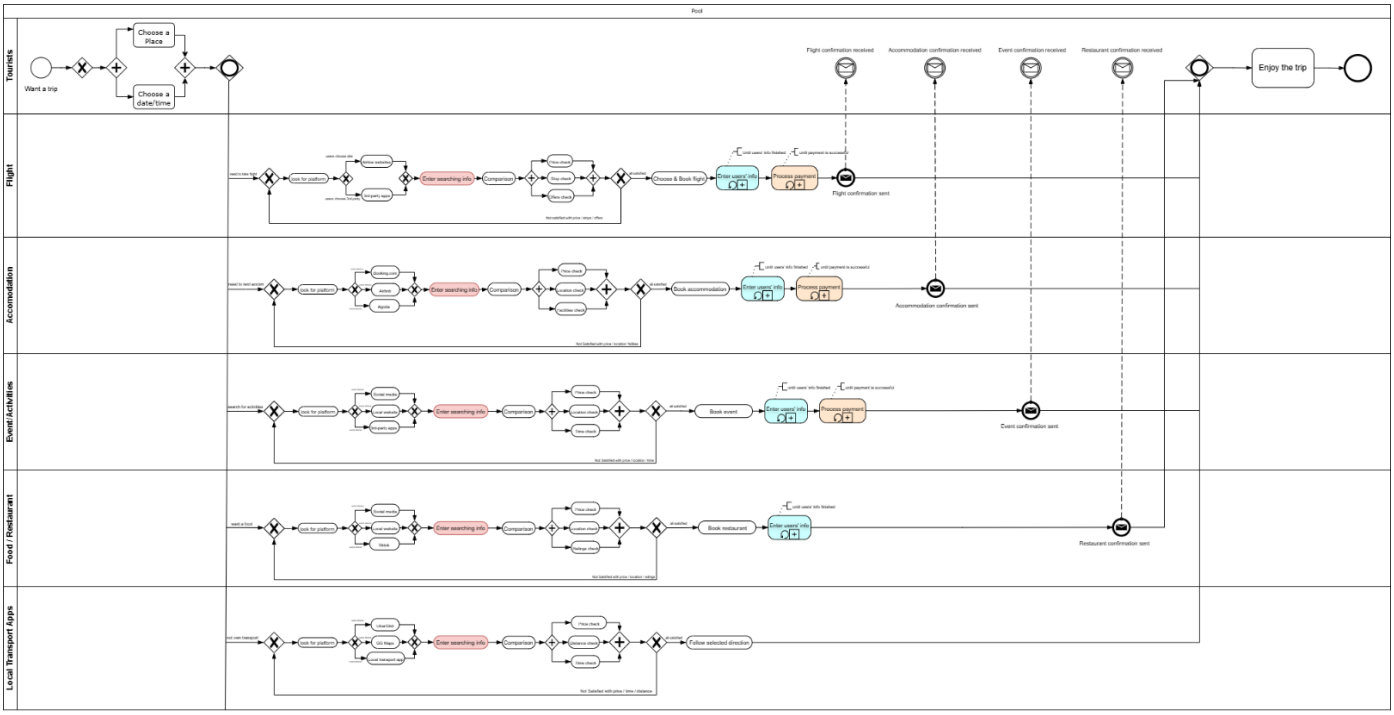


Figure 4. AS-IS process

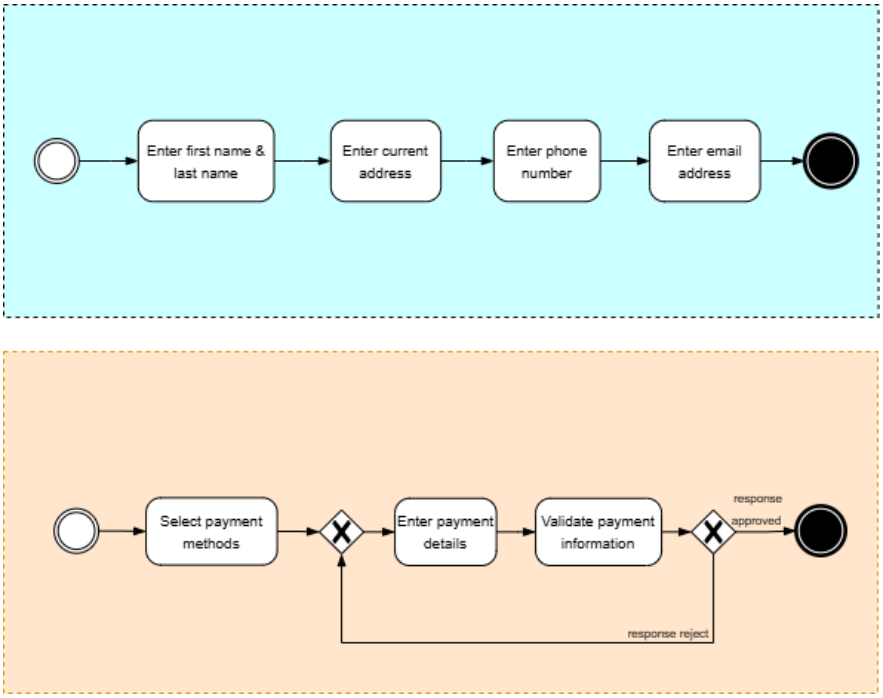


Figure 5. sub-process in AS-IS process

b. TO-BE process

The TO-BE process represents the future state of travel planning through an all-in-one integrated web-based platform designed to overcome the inefficiencies of the current fragmented approach. In this enhanced model, tourists can manage their entire trip — including flights, accommodation, food, activities, transportation, and budget — within a single, seamless application environment. The process begins when users create a trip profile in the app by entering personal details, selecting a destination, and setting travel dates. Once the trip is created, the user's information and preferences are securely stored in the database, enabling personalized recommendations and one-time payment authentication for all bookings. When the payments go through, the confirmations for each booking will be send to the users and store in the database of the current trip.

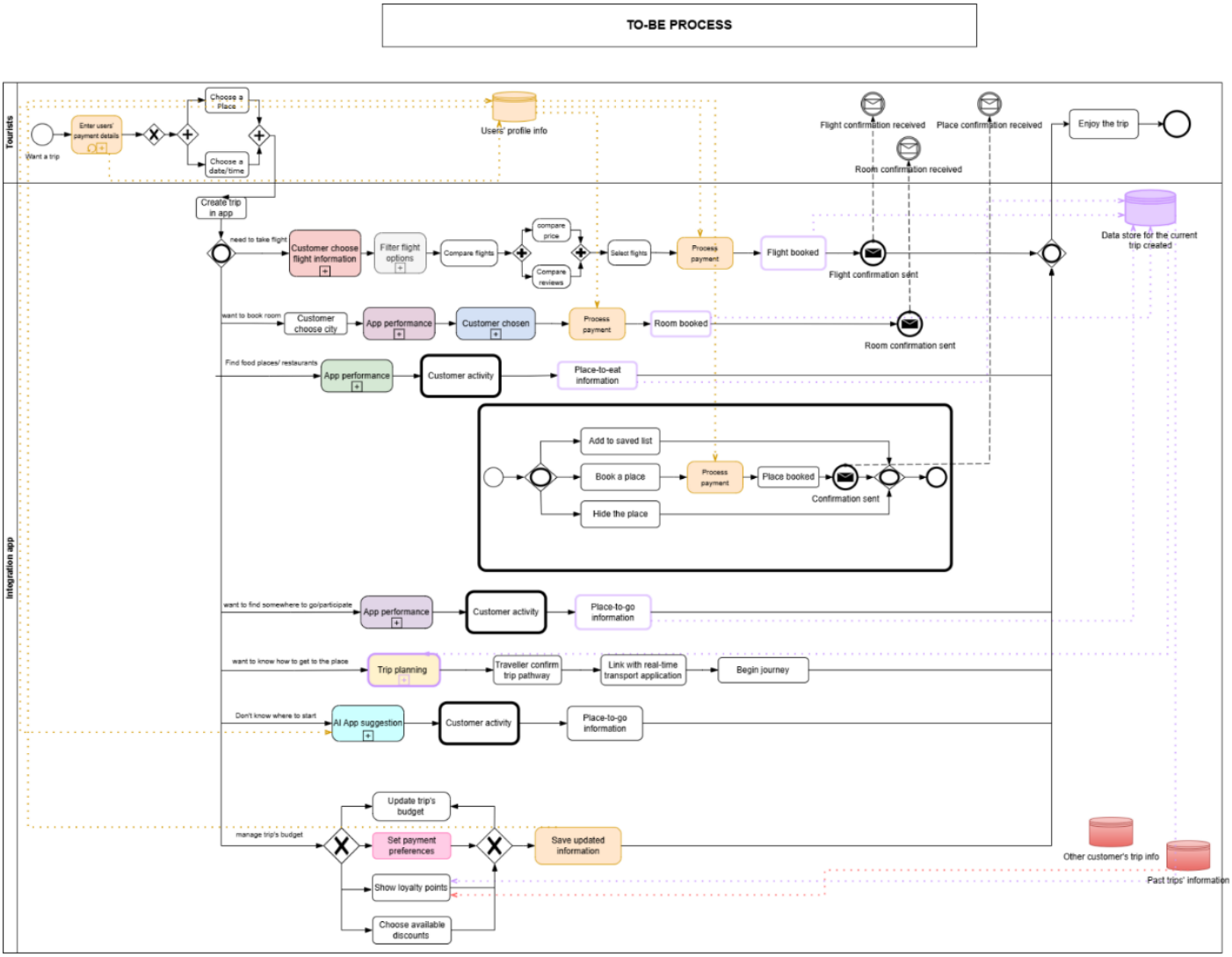
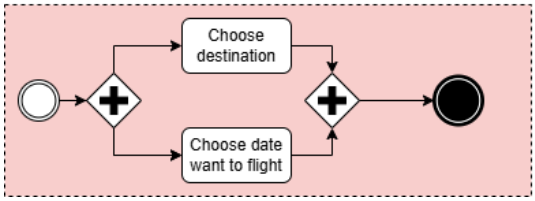


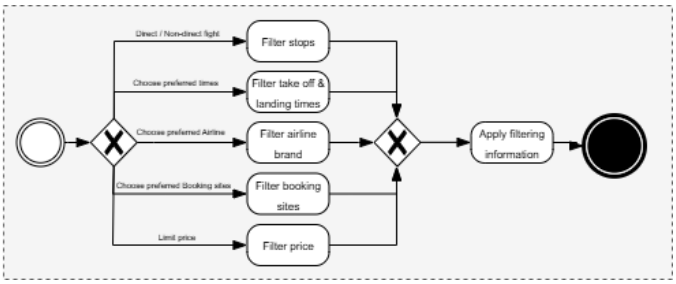
Figure 6. TO-BE process

All travel components are now interconnected. For flights and accommodation, users can filter options, compare prices, and complete payments within the same interface, eliminating the need to switch between apps. Similarly, food and activity planning allows users to explore restaurants, events, or attractions, save favourites, or make instant bookings — all linked to their trip itinerary.

Transportation is integrated with real-time navigation through Google Maps, enabling users to confirm destinations and receive intelligent route suggestions for buses, trains, ferries, or rideshares. Additionally, an AI suggestion feature assists uncertain travellers by analysing their previous trips and preferences to generate a customised plan based on travel type (e.g., family, adventure, or relaxation).

Budget management and loyalty systems are embedded within the same interface, allowing users to track spending, manage rewards, and set payment preferences with options like “Pay Later,” in-wallet, or card transactions.





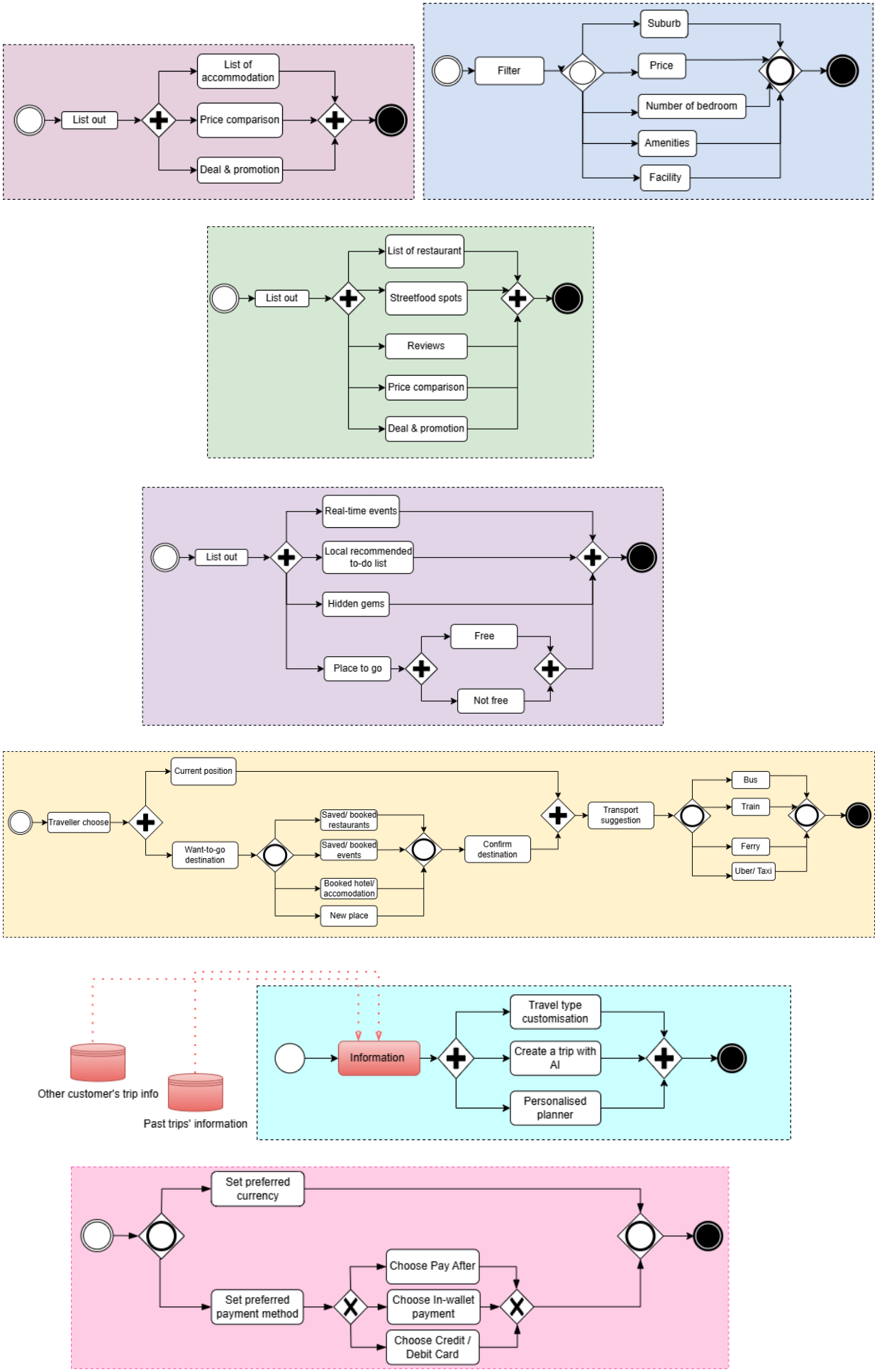


Figure 7. Sub-processes in TO-BE process

3.2.2. High-fidelity prototype - Web-based service

The high-fidelity prototype, developed in Figma, presents a fully functional web-based travel planning platform designed to address key pain points identified in the AS-IS process, including fragmented systems, repetitive app-switching, and disorganised trip management. Acting as a digital travel assistant, the platform enables users to plan, book, and manage every aspect of their journey within a single, seamless interface, reducing cognitive load and enhancing efficiency.

Upon login, users can create a new trip manually or via an AI-assisted generator. By entering destination, travel dates, and purpose, the system constructs a personalised trip dashboard consolidating all operations. Five key modules—Flights, Accommodation, Food & Drink, Places-to-Go, and Events—offer three interaction modes: Saved for bookmarks, For You for curated deals, and AI Recommendations for data-driven suggestions based on user preferences.

Flight and accommodation bookings are integrated with trusted providers, featuring real-time availability, comparisons, and automatic confirmations. The Places-to-Go module highlights both popular and authentic local attractions, while Events uncovers hidden festivals and activities, promoting cultural immersion. Transport integration via Google Maps ensures seamless mobility, and the My Plan section provides Booked and Overview modes for itinerary management.

A Budget Management and Loyalty System streamline payments, applies real-time currency conversion, and rewards engagement. Overall, the prototype delivers a cohesive, intelligent, and user-centred solution, transforming travel planning through integration, personalisation, and automation.

3.3. Business Model

TripMate operates as a multi-sided platform connecting travellers, local service providers, and travel partners within a unified digital ecosystem. Travellers can plan, book, and manage entire trips, while hotels, restaurants, and event organisers promote services and offer exclusive deals. Revenue streams include booking commissions, business subscriptions, advertising, and affiliate marketing. A loyalty system rewards users for bookings, reviews, and social sharing, enhancing retention and lifetime value. Differentiated through integration and AI-driven personalisation, TripMate consolidates all travel components into one interface, reduces decision fatigue, provides tailored recommendations, and leverages strategic partnerships to deliver exclusive discounts, improving efficiency, satisfaction, and local engagement.

3.3.1. Revenue Model

TripMate employs a multi-sided platform (MSP) revenue model, facilitating value exchange among travellers, local service providers, and event organisers. This approach leverages network effects and diversified monetisation strategies to support scalability and long-term sustainability (Parker, Van Alstyne & Choudary, 2016; Bivona et al., 2020). The primary income source comes from transaction-based commissions on bookings for flights, accommodation, dining, and events. For example, a 12% commission on a $100 booking generates $12 in revenue, providing a predictable and scalable revenue base (Evans & Schmalensee, 2015).

Secondary streams include sponsored listings and targeted advertisements from local vendors and tourism partners. Leveraging geolocation targeting and behavioural analytics, these ads convert user engagement into business value (Reinhold et al., 2022). Premium subscriptions offer enhanced features such as ad-free browsing, offline itineraries, and early access to exclusive deals. With an estimated 10,000 monthly users and an ARPU of $2, annual revenue could reach $240,000. Overall, projected revenue distribution is commissions (60%), subscriptions (20%), advertising and affiliates (15%), and loyalty partnerships (5%), supporting TripMate's growth and financial resilience.

3.3.2. Joint Value Proposition

TripMate's joint value proposition is built on value co-creation, following service-dominant logic (S-D logic) and smart tourism principles (Vargo & Lusch, 2008; Chuang et al., 2023). For travellers, the platform provides a seamless, integrated interface to plan, book, and manage trips, consolidating flights, accommodation, dining, and events. AI-driven personalisation and real-time, location-aware

recommendations reduce cognitive load while offering curated access to authentic local experiences often overlooked by traditional travel tools. Centralised bookings and payments save time and minimise errors, enhancing overall satisfaction.

For local businesses, airlines, hotels, and event organisers, TripMate functions as a data-driven marketing channel to reach targeted audiences. Vendors can list services, promote special offers, and analyse customer engagement to optimise marketing strategies. Reward-based loyalty programs increase repeat bookings and user retention. AI-assisted customer support reduces operational costs by 30–40% compared to competitors. By fostering collaboration between travellers and service providers, TripMate strengthens satisfaction, loyalty, and sustainable growth across the tourism ecosystem.

3.3.3. Practical Value to stakeholders

The web-based TripMate platform delivers significant practical value for both travellers and industry stakeholders by addressing inefficiencies in modern travel planning. For travellers, it offers an integrated, all-in-one solution that streamlines booking, navigation, and itinerary management, reducing the need to switch between multiple apps. Centralising flights, accommodation, attractions, events, and transport in a single platform saves time, improves accuracy, and enhances convenience. The AI-driven recommendation engine personalises suggestions based on user preferences and past behaviours, boosting decision confidence, satisfaction, and engagement with local experiences.

For local businesses, hotels, and event organisers, TripMate functions as a targeted digital marketing channel, increasing visibility and customer engagement through listings, promotions, and AI-powered recommendations. By highlighting regional attractions and cultural events often overlooked by mainstream platforms, the system supports local economic growth. Integrated loyalty programs and AI-assisted customer support encourage repeat usage, reduce operational costs, and create a sustainable, collaborative ecosystem that enhances efficiency, connectivity, and shared value across all stakeholders

4. Conclusion

This project successfully designed TripMate, an All-in-One Tourism Journey App aimed at reducing travellers' stress and simplifying trip management by integrating planning, booking, navigation, and social sharing within a single, user-focused platform. By applying the design thinking methodology—empathise, define, and ideate—the project gained deep insights into traveller needs, identified key pain points, and generated innovative solutions that informed the development of a high-fidelity Figma prototype.

The prototype demonstrates an interconnected ecosystem where AI-driven recommendations, personalised itineraries, and seamless integration of flights, accommodation, events, and local experiences enhance convenience, safety, and engagement. It shows potential to improve trip planning efficiency, elevate user satisfaction, and provide local businesses and event organisers with an effective digital marketing channel.

Utilising the Business Process Management (BPM) lifecycle ensures systematic implementation, continuous optimisation, and scalability. Overall, TripMate offers a practical, intelligent, and sustainable solution that transforms fragmented travel planning into a seamless, enriching, and personalised experience for all stakeholders.

5. Recommendations

To advance this concept toward practical implementation, several recommendations are proposed. First, extensive usability testing with diverse traveller profiles is essential to refine the interface, ensuring intuitive navigation and a seamless user experience. Strategic partnerships with established platforms such as Booking.com, Google Maps, and Eventbrite should be pursued to enhance data accuracy, reliability, and real-time integration. The AI-powered recommendation system can be optimised further to deliver personalised, context-aware suggestions based on user preferences, past behaviour, and travel trends.

Additionally, the app should prioritise user trust by implementing robust data privacy measures and adhering to GDPR and ISO standards. Incorporating gamification, social sharing features, and accessibility options—including multilingual support—will encourage engagement and inclusivity. Collectively, these

recommendations support the development of a scalable, user-centred, and effective digital travel ecosystem.

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Appendix

https://app.diagrams.net/#G1vh1-UYUd7BDLPfQ2cZxpLbIVDtD27DTA#%7B%22pageId%22%3A%22c7488fd3-1785-93aa-aadb-54a6760d102a%22%7D
Link for AS-IS ad TO-BE process

https://www.figma.com/design/Xj3mvFItmpgmtM5u1MLVze/lynndao-and-her-mess?node-id=6-1655&p=f&m=draw
Link for high-fidelity prototype